Audit, Freeze and Experiment Record

Financial Fraud Detection in Mobile Money Networks Using Supervised Machine Learning

Condensed record, October 2026

About this document

The project's working notes (15 Markdown files, about 3,500 lines) record how the study's measurement system was audited, repaired and frozen, and what each later experiment found. This document condenses them into one readable record. Every number here is copied from those notes or from the result files they cite; nothing was recomputed for this summary. File names in `typewriter` refer to the repository (`results/`, `src/`, `scripts/`).

Two corrections that later checks found in the notes are applied here and marked [**corrected**].

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1 Data and the leakage finding

- **Dataset.** PaySim (`ealaxi/paysim1`), 6,362,620 rows, 8,213 frauds (0.129%), 743 hourly steps (about 31 days). Its SHA-256 is checked on every load.
- **Simulator rule.** A three-clause boolean rule over the origin-balance columns recovers the fraud label at precision 0.99988 and recall 0.97699. This is an in-sample description of the generator, not a held-out detector.
- **Single-feature leak.** `errorBalanceOrig` alone reaches AUC 0.9020.
- **Drain signature.** The condition `amount == oldbalanceOrg` holds for 97.82% of frauds and 0.0013% of legitimate transactions, with a median gap of exactly 0.0.

- **Fraud types.** Fraud occurs only in TRANSFER and CASH_OUT, by construction.

Decision. The main experiments use the `no_origin_balance` feature set (14 features). It drops `errorBalanceOrig`, `errorBalanceDest`, `oldbalanceOrg`, `newbalanceOrig`, `origZeroBefore` and `origZeroAfter`. It was selected on validation NER (0.1147) by the pre-declared rule: take the least-ablated feature set whose best NER is at least 0.05. The only permitted description of it is "reduced simulator artefact exposure". It is not simulator-independent or artefact-free.

Destination-side check (`destination_artefact_diagnostic.md`). The same argument was tested for the destination balances, and it did not hold:

- The strongest destination feature reaches AUC 0.6237 alone.
- Lifts are only 1.3–1.5×.
- The suspected mechanism, "the receiving account is never credited", occurs for 60% of legitimate and 52% of fraudulent transactions (lift 0.86). It points the wrong way.
- Removing the destination block would cost 0.25–0.43 PR-AUC; logistic regression falls to 0.078.

The destination columns stay in the main feature set, and `no_balance_state` is kept for diagnosis only.

Chronology (CHANGE-01). The ablation declared in notebook 02 removed two features; the one executed removed six. The repository cannot prove whether that choice was fixed before the cost-sensitive results were seen. The selection is therefore described as "criterion-driven and mechanically enforced", never as pre-registered.

2 Audit and repair

An audit of the measurement system found critical (C), major (M) and minor (m) issues (`repair_report.md`). No experiment was redesigned, no hyperparameter tuned and no result deleted. Superseded files remain on disk, versioned, and `results/experiment_index.json` records the status of each.

Issue	Problem and repair	Status
C1	The temporal split realised 95.6/3.0/1.4 (test prevalence 1.391%). Its boundaries were re-derived from row counts (322/377, giving 69.68/15.30/15.02).	Fixed, rerun
C2	A sweep of the decision cost ratio had been read as a training-cost sweep. It was relabelled `06_decision_cost_sweep.csv`.	Fixed (relabel)
C3	The feature set, best configuration and learner had been selected on test NER. They are now selected on validation; the feature set is unchanged.	Fixed, rerun
M1	Overlap of confidence intervals was used as a significance rule. It was replaced by a paired stratified bootstrap, $B = 2000$.	Fixed
M3	The destination-side asymmetry was diagnosed (above); the main feature set is not changed.	Diagnosed
M4	Configuration F trains on $R \cdot s_i$ but is scored on R. It is reclassified as a sensitivity analysis.	Documented
M5	Configurations B and C are the same condition at the default R, so there are six distinct conditions, not seven.	Documented
M7	The notebook 05 feature builder was fitted on the whole subset. The effect was measured as inert.	Fixed
M8	All cost-sensitive results came from one seed and one split.	Closed later by E3
M9	Permutation importance is distorted by correlated features.	Open
M10	The validation set was reused three times during tuning.	Documented

What the repairs changed

- **The temporal result reversed.** The temporal split is now harder than the stratified one for 3 of 4 learners. Fraud prevalence is strongly non-stationary across it (train 0.082%, validation 0.059%, test 0.420%).
- **The cost-sensitive comparison became stronger.** All 12 comparisons showed lower PR-AUC than the unweighted model, and every paired interval excluded zero. The four previously called "indistinguishable" became small negative differences.
- **Best configurations changed for 3 of 4 learners** once selection moved to validation.

3 The frozen baseline

Recorded in `frozen_baseline_protocol.md`. Its machine-readable copy passes 9/9 checks: `results/frozen_experimental_baseline.json`.

- **Split.** Stratified 70/15/15 with split seed 42. The feature builder and any resampling see training rows only. Thresholds are selected on validation, locked, and applied to test once.
- **Cost.** $C_{FP} = 1$, $C_{FN} = R$, with $R = N_{\text{legit}}/N_{\text{fraud}} = 773.70$. The Bayes threshold is $\lambda = 1/(1 + R)$, and $\text{NER} = (\text{FP} + R \cdot \text{FN})/(R \cdot N_{\text{fraud}})$, so flagging nothing scores 1. R_{train} and R_{eval} are always recorded separately.
- **Models.** Logistic regression, decision tree, random forest and XGBoost, all at fixed, untuned defaults. LightGBM is excluded because its fit failed (PR-AUC 0.043).
- **Configurations.**
  - A: unweighted.
  - B and C: unweighted with a Youden or a NER-optimal threshold.
  - D/E: class weighting.
  - G: random under-sampling.
  - F: per-instance severity weighting, a sensitivity variant only.
- **Metrics.** PR-AUC ranks models; NER measures decision risk. Accuracy is never computed. The report writes "the paired interval excludes zero", never "statistically significant".

Configuration F (`config_f_cost_objectives.md`). F trains with weights $R \cdot s_i$, where s_i is the percentile rank of the amount, so $E[s_i] \approx 0.5$. It is then thresholded and scored as if every miss cost R. It therefore carries about half of D's weight and is judged on a different objective. Its result cannot show that severity weighting works or fails.

4 E3: robustness across splits and seeds

`E3_report.md`. The run covered 220 fits: 4 learners × 4 variants × 4 split seeds × 4 model seeds. It omitted 36 deterministic logistic-regression duplicates. All 220 completed, with QC 22/22 and the smoke test 16/16. At split 42 and seed 42 every learner reproduces the frozen baseline exactly.

- **Ranking quality.** All 116 core paired comparisons (weighting and under-sampling) have lower PR-AUC for the cost-sensitive model. Mean differences range from −0.015 (decision tree, weighting) to −0.515 (decision tree, under-sampling). Configuration F, reported separately, is negative in all 52 pairs.
- **Decision risk depends on the learner, consistently.**
  - Logistic regression and the decision tree have lower NER with cost-sensitive training in **52 of 52** pairs [**corrected**]. The note says 64 of 64, but these rows hold 4+16+16+16 = 52 pairs.
  - The random forest and XGBoost have higher NER in 61 of 64 pairs.
  - XGBoost with under-sampling (mean +0.0028) is indistinguishable from no effect.
- **Variability.**
  - The split moves PR-AUC more than the training seed does: split $\sigma \approx 0.010$–0.016 against seed $\sigma \approx 0.001$–0.003.
  - The exception is the decision tree with under-sampling (seed $\sigma = 0.042$).
  - Split seed 42, used for every earlier result, scores lowest of the four for every learner.

5 E2: sweep of the training cost ratio

`E2_report.md`. Seven R_{train} levels: 1 (unweighted), 10, 50, 100, 773.70, 2000 and 5000. That gives 364 cells (260 new, 104 reused from E3, verified identical), with R_{eval} fixed at 773.70 and QC 21/21.

- **PR-AUC** falls with R_{train} across the weighted levels for all four learners. Including $R = 1$, the fall is monotone only for the random forest and logistic regression; XGBoost and the decision tree peak at $R = 10$. The fall holds for the means, not for every cell.
- **NER** has no common direction. It rises with R_{train} for XGBoost and the random forest, falls for logistic regression, and is flat for the decision tree. The lowest observed NER is at $R = 1$ (XGBoost), 10 (random forest), 773.70 (decision tree) and 2000 (logistic regression). No level is described as optimal.
- **Learner ranking.** The order XGBoost > random forest > decision tree > logistic regression holds in all 112 matched cells at every level.
- **Variance claim corrected in the note.** The split exceeds the seed by 3–13× for XGBoost and the random forest, not by "two orders of magnitude".

6 E4: changing prevalence alone

`E4_report.md`. Test negatives were subsampled while all 1,232 frauds were kept. The model and threshold stay fixed and nothing is refitted. The run covered 104 cells and 62,504 evaluation rows at 7 prevalence levels from natural (0.129%) to balanced (50%), with QC 14/14.

- **Precision** rises 21.8–57.0× (weighted arm) from natural to balanced prevalence, while recall does not change. The "precision collapse" is fully explained by the prevalence evaluated.
- **PR-AUC is not prevalence-invariant.** Logistic regression rises from 0.158 to 0.971 with an unchanged model.
- **NER** changes least (−22% to −35%). At balanced prevalence it tends to 1 − recall.
- **Balanced evaluation hides learner differences.** In the unweighted arm at 50%, the full learner order survives in 5.8% of cells.
- **A bug was caught.** The reproduction check found transposed seeds in a first draft that the smoke test had missed.

7 E6: where the two threshold rules diverge

`E6_report.md`. The Youden rule (B) and the NER-optimal rule (C) were compared at 8 evaluation cost ratios. The ratios run from 1/64 to 64 times the class ratio, plus the validation ratio 773.67. The run covered 104 cells and 832 paired comparisons, with QC 15/15. The ratio grid was chosen by the analyst, not pre-declared.

- **At the class ratio the two rules are one rule.** They chose the same threshold in 104 of 104 cells, with a gap of exactly 0, as the derivation predicts.
- **Rule C never loses.** The smallest ΔNER over all 832 pairs is exactly 0.
- **Below the class ratio, Youden can be worse than flagging nothing.** NER exceeds 1 in 100 of 832 rows, with a peak of 6.38.
- **Above it, Youden stalls.** It plateaus at NER ≈ 0.107, while C reaches 0.0092.
- **The literature disagreement dissolves.** The reported contrast between a "statistical" and a "business" threshold is a distance between two cost ratios.

8 Authentication as a design contribution

`authentication_design_contribution.md` (CHANGE-06). PaySim has no session, device, SIM, PIN, channel or location field, so the authentication research question cannot be answered empirically. It is answered as a design.

The design. The risk bands are cut at $\lambda/10$, λ and 10λ, so one R sets both the alert threshold and the authentication ladder. Each band has its own action and requirement:

- **Low:** allow.
- **Medium:** monitor.
- **High:** step-up with a biometric factor.
- **Critical:** block and verify by a separate channel.

What is measured is friction only. For XGBoost on the test split, 97.04% of transactions are never escalated, while the 2.96% that are escalated carry 90.2% of the fraud.

Not claimed. No control was tested, the usability cost of false rejects is unmeasured, and PaySim fraud drains accounts rather than compromising authentication. The biometric step also excludes feature-phone (USSD) users.

9 Claim strength and literature

Claim categories (`claim_taxonomy.md`). Every claim is placed in one of five categories:

1. demonstrated mathematically, e.g. the Youden and NER-optimal thresholds coincide at the class ratio;
2. derived and confirmed in measurement, e.g. E6's 104 of 104 agreement;
3. measured on PaySim, always with its scope stated;
4. associated, with the confound named, e.g. the feature-importance ranking while M9 is open;
5. cannot be claimed from PaySim, e.g. that authentication controls work, or transfer to real networks.

The claim lists in the frozen baseline were updated after E2–E6 (CHANGE-07), without changing any number.

Closest prior work (`literature_note_hajek_2023.md`, `hajek_comparability.md`). Hajek, Abedin and Sivara-

jah (2023) found that under-sampling with XGBoost gave the best cost saving on PaySim, the opposite of this study's ranking result. Four explanations are ranked as hypotheses; the first is that they may have kept the origin-balance columns. The full text has not been obtained, so the comparison stays open.

Drafting notes. `report_outline.md` and `report_review.md` are planning and review notes for an earlier draft of the report. They contain no results of their own and are superseded by the current revision.

10 Known limitation of the bootstrap intervals

The paired bootstrap (`evaluate.paired_bootstrap_pr_auc`) computes PR-AUC with its own routine, which does not group tied scores. The reported absolute PR-AUC values use scikit-learn, which does. For the decision tree, whose scores take only 29–75 distinct values, the two can differ by up to 0.033.

A read-only check (23 October 2026) replayed the identical 2,000 resamples, which reproduced the frozen intervals exactly, and recomputed each comparison with tie-aware PR-AUC:

- All 12 intervals still exclude zero, and no comparison changes direction.
- The largest change is for the decision tree with under-sampling: $\Delta = -0.536$ instead of -0.569.
- The largest crossing count rises from 3 to 10 of 2,000 replicates.

The frozen results are unchanged and remain the primary analysis.

11 What remains open

- **M9:** grouped permutation importance was not run.
- **Configuration F:** no example-dependent experiment, consistent from training to scoring, was run.
- **Thresholds under prevalence shift:** re-selecting the threshold at a new prevalence was not tested.
- **Hajek et al.:** the full-text comparison is still to be made.
- **Scope:** one simulated dataset, one feature set and four stratified splits. No claim transfers to real mobile money networks.